

Classification of enzymes

Enzymes are classified depending on the type of reaction they catalyze. The following are some of the classes of enzymes.

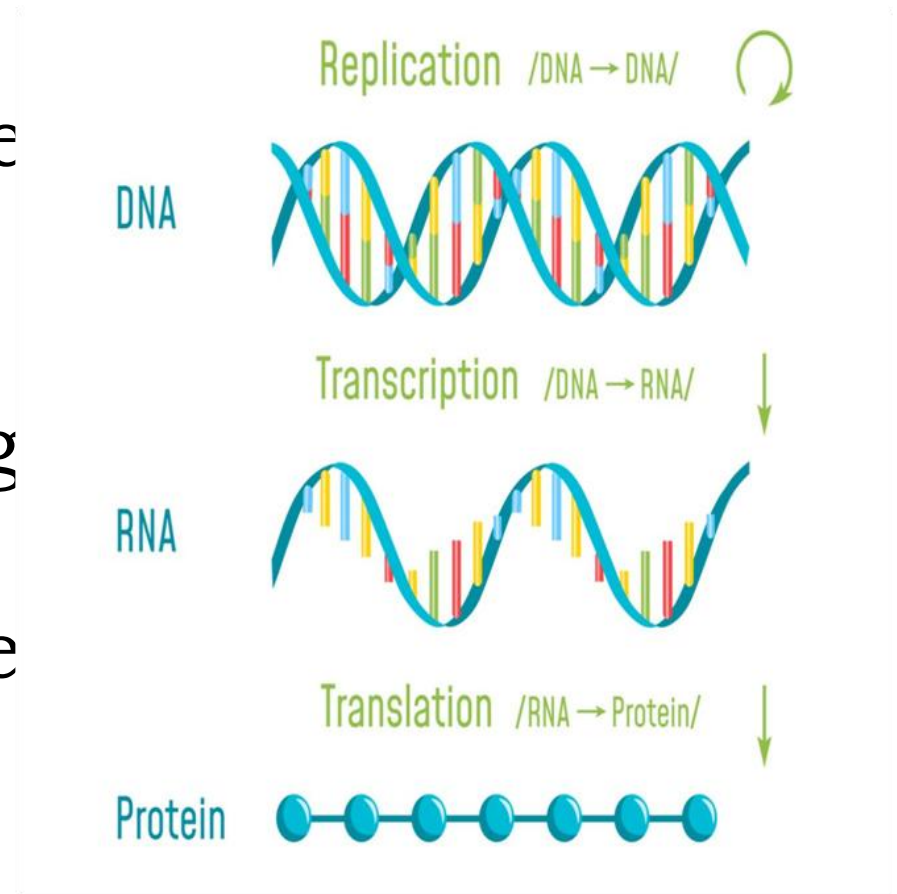
- 1. Isomerase;** these catalyze reactions involving isomerism
- 2. Phosphorylases;** these catalyze reactions involving addition of a phosphate
- 3. Hydrogenases;** these catalyze reactions involving addition of hydrogen.
- 4. Dehydrogenase;** these catalyze reactions involving removal of hydrogen.
- 5. Kinases;** these catalyze reactions involving movement of molecules from one area to another.
- 6. Carboxylases;** these catalyze reactions involving addition of Carbon dioxide.

Intracellular and extracellular enzymes

*Enzyme can also be described as being **intracellular or extracellular.***

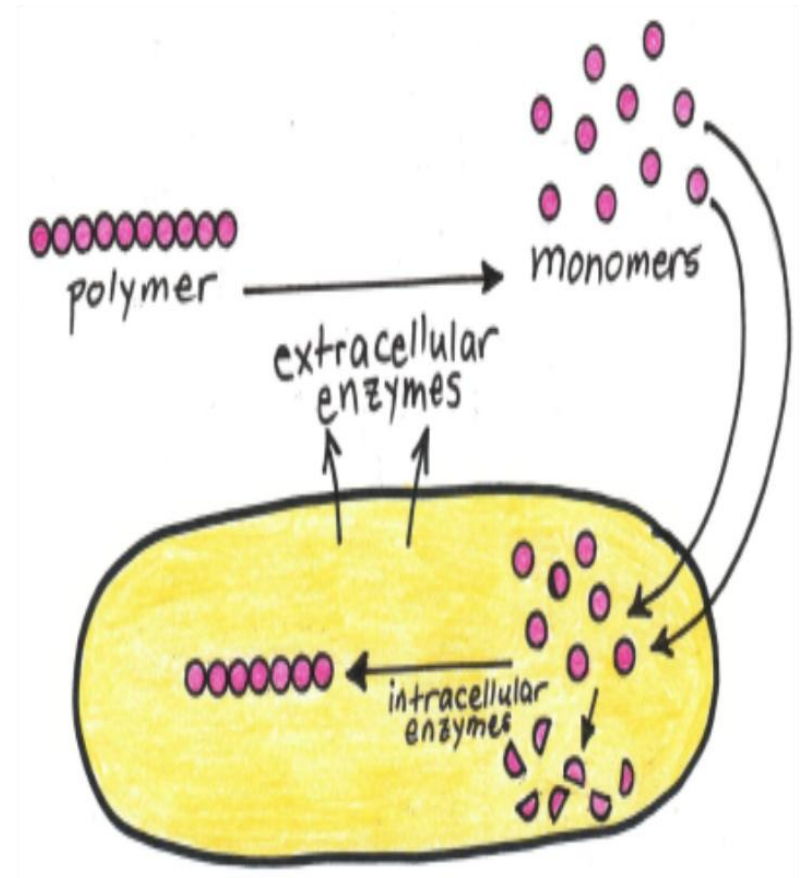
Intracellular enzymes

Intracellular enzymes are those which catalyze reactions inside the cells producing them, e.g. all respiratory enzyme are intracellular.



Extracellular enzymes

Extracellular enzymes are those produced by a cell to catalyze reactions outside that cell. All digestive enzymes in man are extracellular.



Nomenclature of enzymes

- ❖ Enzymes are named by adding a suffix “**ase**” to their substrates.
- ❖ **A substrate** is a substance, which the enzyme acts upon, or simply it is the *raw material* for the enzyme.

Examples of enzymes and their substrates

Enzyme	Substrate
Peptidase	Peptides
Lipase	Lipids
Maltase	Maltose
Sucrase	Sucrose
Lactase	Lactose
Cellulase	Cellulose

Nomenclature of enzymes

- Some enzymes however retained their names they had before this convention. Such enzymes include **pepsin** and **trypsin**.
- Sometimes the enzymes digesting carbohydrates are generally called ***carbohydrases*** and those digesting proteins as ***proteases***.

PROPERTIES OF ENZYMES

- ❖ They are all **protein** in nature.
- ❖ They are **specific** in their action i.e. they catalyze specific food i.e. Maltase on Maltose.
- ❖ They speed up the rate of chemical reactions **(they are catalysts)**.
- ❖ They are **effective** even in small amounts.
- ❖ They **remain unchanged** at the end of the reaction.

PROPERTIES OF ENZYMES

- ❖ They are ***denatured*** by ***high temperatures*** since they are protein in nature and are inactivated by low temperatures.
- ❖ They are ***inactivated*** by ***inhibitor*** chemicals (poisons e.g. cyanide).
- ❖ They work at ***a specific PH***. (either acidic or alkaline).
- ❖ Their reactions are ***reversible***.
- ❖ Their activity can be ***enhanced*** by ***enzyme activators*** e.g. chloride ions activate amylase.

FACTORS AFFECTING ENZYME ACTIVITIES

To investigate the effects of a given factor on the rate of enzyme controlled reactions, all other factors should be kept constant and at optimum levels so as to obtain accurate results.

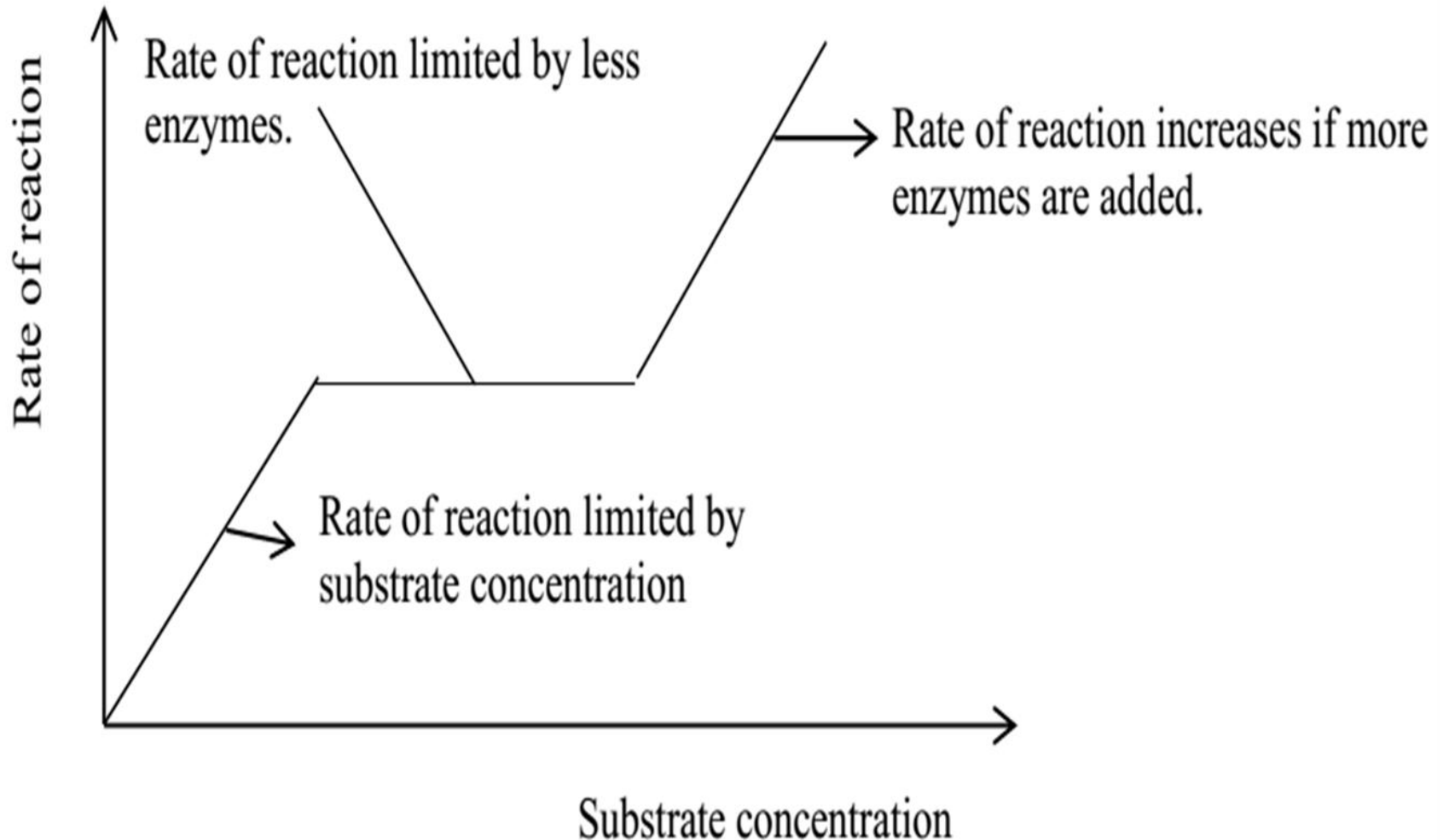
The factors are:

- 1. Temperature*
- 2. Concentration of the substrate*
- 3. PH of the medium*
- 4. Presence of activators*
- 5. Presence of inhibitors*
- 6. Concentration of the enzyme*

1. Concentration of substrate:

- ❖ A substrate is a substance (food) acted upon by the enzyme to form simpler products.
- ❖ *The rate of enzyme reaction increases with increase in substrate concentration and enzymes work slower when the substrate concentrations low.*
- ❖ However, further increase in substrate concentration will not increase enzyme reaction rate since all its active sites are fully saturated with food.

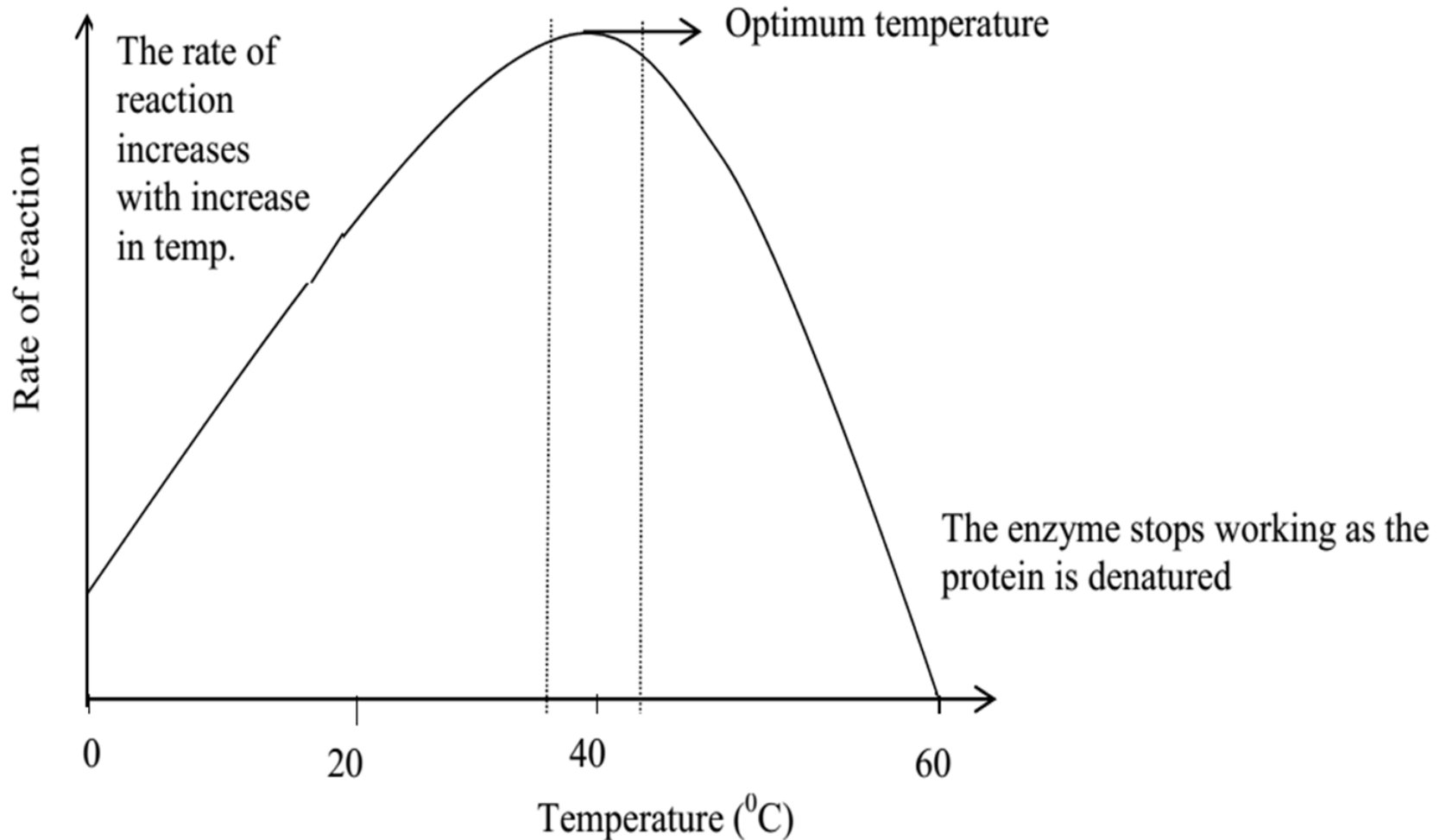
A graph showing how the rate of reaction varies with substrate concentration



2. Temperature:

- ❖ Enzymes work best at **optimum temperatures** of (approximately 37 °C). At very low temperatures, the rate of enzyme reaction is very slow because the enzyme is inactive at such low temperatures.
- ❖ *As the temperatures increase, the rate of reaction also increases gradually until it attains a peak where it has maximum activity and this always correspond at optimal temperatures.*
- ❖ **An optimal temperature is which promotes maximum enzyme activity.** However with further increase in temperature, the rate of reaction decreases exponentially, sharply, steeply since at high temperatures, the enzyme is denatured ie the active site of the enzyme which is (protein in nature) is a altered (changed) or completely destroyed.

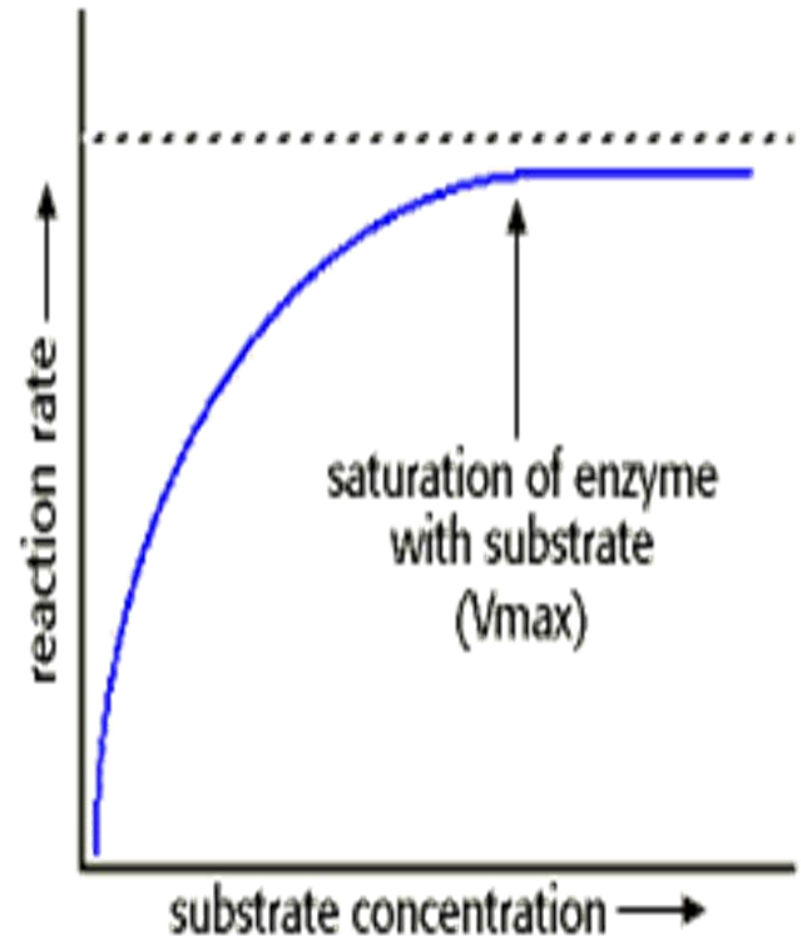
A graph showing the variation of enzyme activity with temperature



Graph showing effect of enzyme concentration on enzyme activity

3. Enzyme concentration:

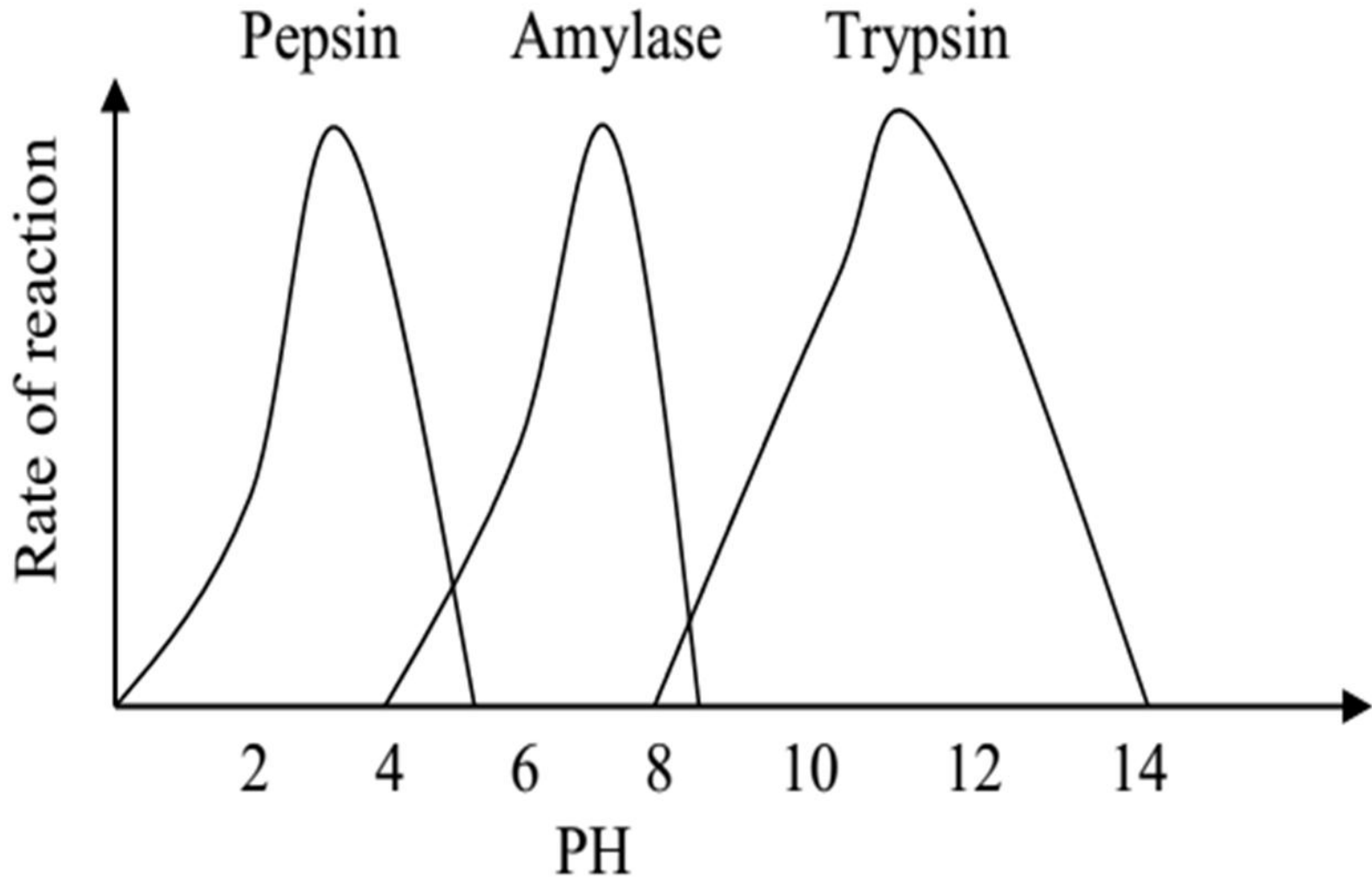
- As the concentration of the enzymes increases, the rate of reaction also increases until all the substrates are being acted upon when the rate finally becomes constant.



4. The PH of the medium.

- ❖ *Enzyme reactivity is reduced or stopped completely if placed in a medium whose PH is different from that in which it works best (optimum PH).*
- ❖ **PH varies** slightly above or below an enzyme's optimum PH resulting in a marked fall in the enzyme efficiency. E.g. pepsin enzyme in the human stomach has a maximum activity with in acidic pH of 1.5 and 2.5 while the enzymes in the duodenum e.g. trypsin work at maximum with in alkaline pH of 8.5 to 9.5.

A graph showing variation of different enzyme activity with PH



5. Presence of enzyme inhibitors

Enzyme activities decrease in presence of enzyme inhibitors and increase in their absence.

6. Presence of activators

*Enzyme activators **increase** with presence of enzyme activators and **decrease** with absence of enzyme activators.*